

Power Plants, Counties, and PM_{2.5}

A Spatial Overlay Brief

Ali Hunter

June 2, 2026

Overview

This brief summarizes a spatial analysis linking U.S. power plant locations to county-level air quality across the contiguous United States. Using three overlay techniques — point-in-polygon joins, buffer intersections, and raster zonal statistics — I ask whether counties with power plants nearby tend to have higher mean PM_{2.5} exposure than those without. The data are EIA-860 generating facilities (every generator ≥ 1 MW), U.S. Census county boundaries, and an annual mean PM_{2.5} surface at roughly 5 km resolution.

Methods

All vector layers were reprojected to Conus Albers Equal Area (EPSG:5070) before any spatial operations to ensure accurate distance and area calculations.

Q1 — Plants per county used a point-in-polygon spatial join (`st_join` with `st_within`). Each plant point was matched to its containing county, counts were aggregated to the county level, and unmatched counties received zero. A small number of facilities (24) fell outside any county boundary — likely coastal or boundary-straddling geometries — and were dropped.

Q2 — Plants within 25 km used a buffer-intersect approach. Each plant was buffered by 25 km and counties were flagged if they intersected any buffer (`st_intersects`). The 25 km threshold reflects the regional scale over which PM_{2.5} disperses — a plant just outside a county border is still a meaningful exposure source — while staying tight enough to exclude truly distant facilities.

Q3 — Mean PM_{2.5} used area-weighted zonal statistics via `exactextractr::exact_extract`, which weights each raster cell by the fraction of its area falling inside the county polygon. This is more accurate than centroid-based approaches, especially for small or oddly shaped counties.

Results

Q1: Power Plant Distribution by County

Powerplant facilities per county

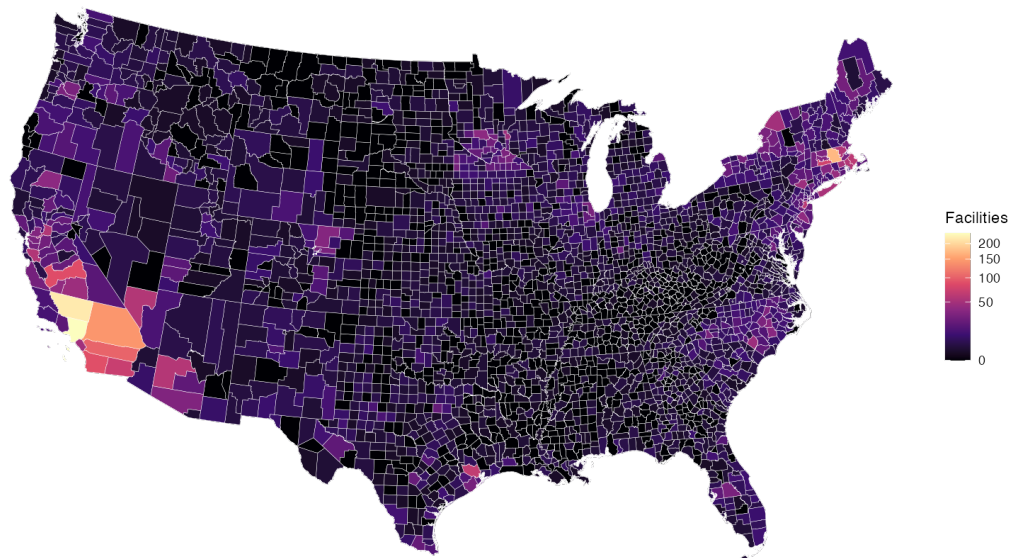


Figure 1: Raw facility count per county (square-root color scale).

Most counties have few or no power plants. Facility counts are heavily right-skewed, with hotspots concentrated in the Northeast corridor, the Los Angeles basin, and parts of the South. The square-root color scale makes this clearer — the handful of counties with 100+ facilities would otherwise swamp the rest of the map.

Normalizing by area shifts the picture. The Northeast looks even more concentrated on a per-km² basis, while large Western counties — which have high raw counts spread over enormous areas — fade out. Density is arguably the more meaningful measure for thinking about local exposure.

Q2: Counties Near a Power Plant

At 25 km, the vast majority of counties in the contiguous U.S. are flagged as near a power plant — the map is almost entirely orange, with only scattered rural counties in the Great Plains and Mountain West coming up empty. This is a useful reality check: power plants are genuinely ubiquitous at this scale. Dropping to 1 km reveals much more spatial variation, with large swaths of the rural West and Midwest now excluded. The choice of threshold matters a lot for how the “nearby” comparison will read.

Powerplant facilities per 100km²

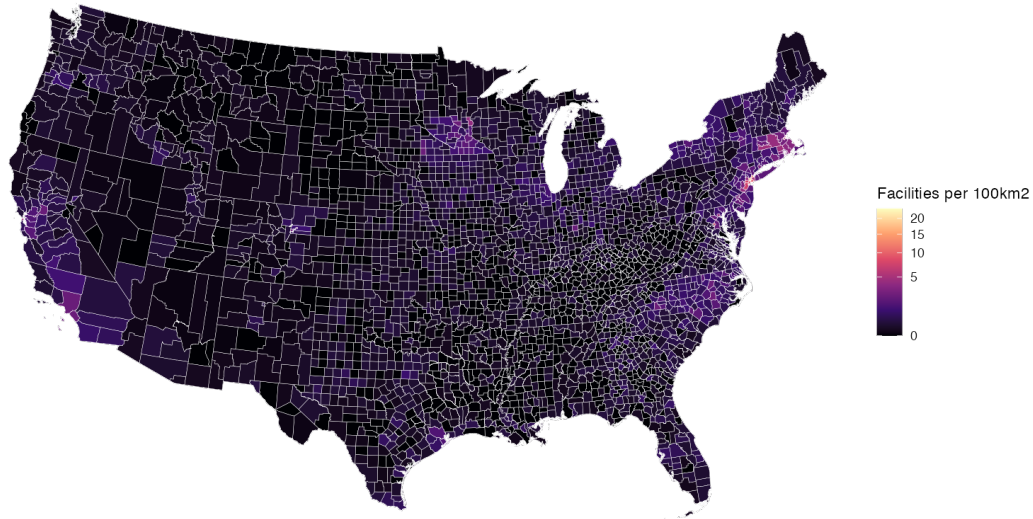


Figure 2: Facility density per 100 km². Normalizing by area shifts the story toward the Northeast, where counties are small and densely packed.

Q3: Mean PM_{2.5} by County

PM_{2.5} is highest in California (especially the Central Valley and the LA basin), the Mid-Atlantic and Northeast corridor, and parts of the industrial Midwest and Southeast. The West is generally cleaner except for California. This spatial pattern reflects a mix of topography (California's valleys trap pollution), population and industrial density (the Northeast), and agricultural burning.

Under the current EPA annual standard of 9 $\mu\text{g}/\text{m}^3$, noncompliant counties cluster heavily in California, the Northeast, and scattered pockets of the South. The geographic concentration is striking — this is not a diffuse national problem but a regional one.

Mapping the *share* of county area above 9 $\mu\text{g}/\text{m}^3$ reinforces this: the California hotspots are not just exceeding the threshold on average, but exceeding it across essentially the entire county area. Counties in the Northeast exceed it in a large fraction of their area as well.

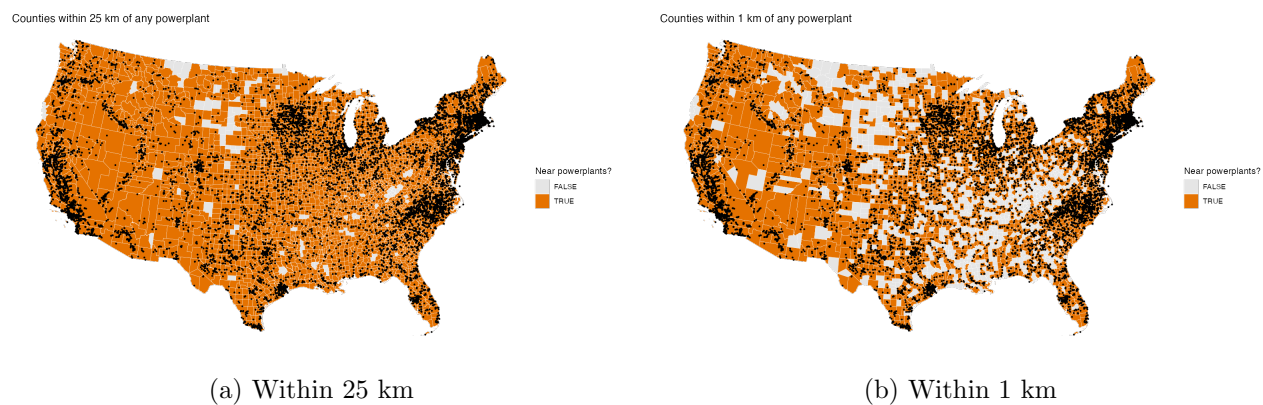


Figure 3: Counties flagged as “near” a power plant at two buffer distances.

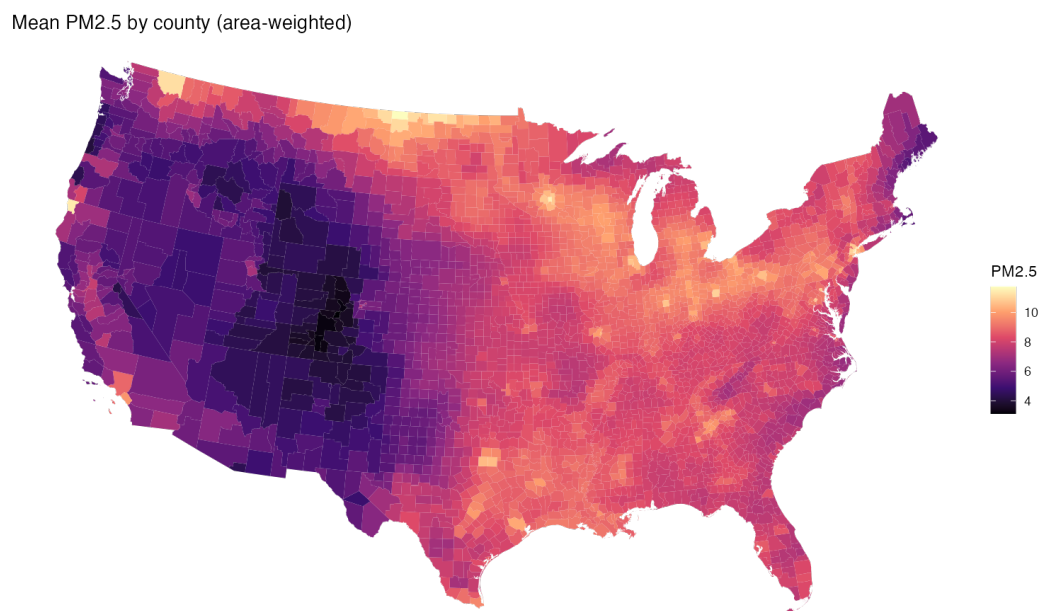


Figure 4: Area-weighted mean PM_{2.5} ($\mu\text{g}/\text{m}^3$) per county.

Counties whose mean PM_{2.5} exceeds 9 ug/m³

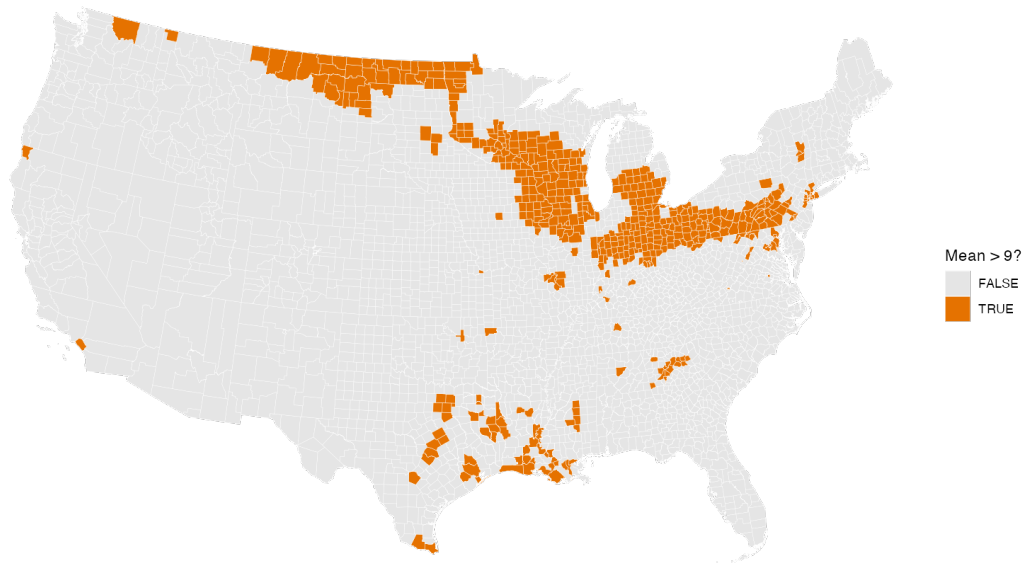


Figure 5: Counties whose mean PM_{2.5} exceeds 9 $\mu\text{g}/\text{m}^3$ — the current EPA NAAQS annual standard.

Share of county area with PM_{2.5} > 9 ug/m³

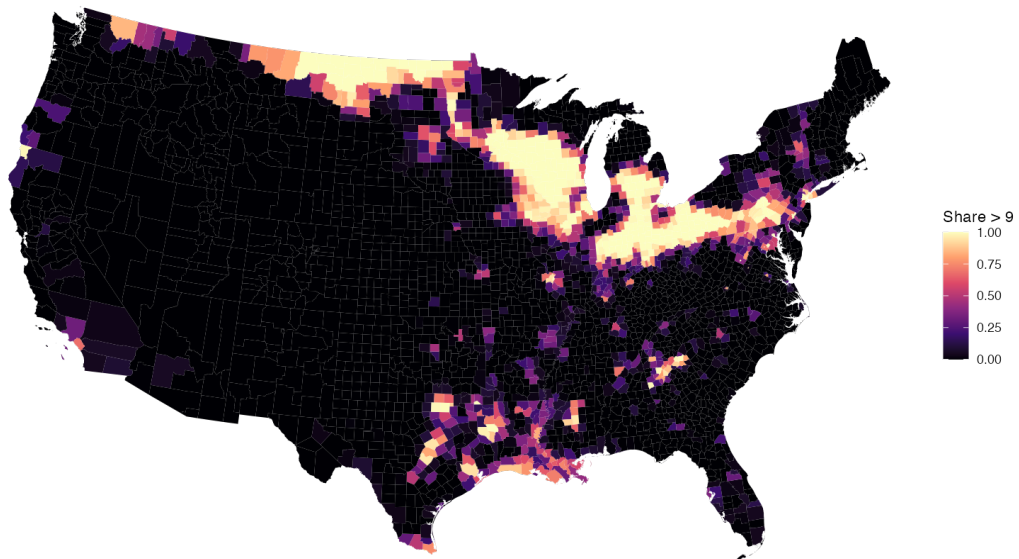


Figure 6: Share of each county's area where PM_{2.5} exceeds 9 $\mu\text{g}/\text{m}^3$.